Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

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Abstract

When language models disagree on a question, the disagreement is often read as missing knowledge. If so, questions that need the same knowledge should be answered right or wrong together. We look for this structure with the eigenvalues of the correlation matrix between questions. Within one model, independently sampled answers show none that we can detect, but structure appears when all questions in one pass over the benchmark share a random seed or few-shot examples, drawn anew for each pass. Across 24 models, the largest structure beyond overall skill is a preference for one answer letter, mostly from one 1B-parameter model; when the options are rotated, it follows the letter far more than the question. Counting a question correct only if it is right in every rotation removes this structure, at least for a preference that acts on guesses. A marginal excess remains, mostly from two pairs of near-duplicate models, and a second skill half as strong as overall skill would usually be missed. Before reading disagreement as missing knowledge, sample independently and rotate the options.

1. Introduction

Self-consistency samples a language model several times on the same question and reads disagreement among the answers as uncertainty about the answer (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). In a deep ensemble, disagreement among the members is read as epistemic uncertainty: uncertainty that comes from missing knowledge, as opposed to noise that would remain even if the model were known (Lakshminarayanan et al., 2017; Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). Recent scores add disagreement between different models to self-consistency as exactly such an epistemic term (Hamidieh et al., 2026).

This reading makes a prediction. A model that lacks some piece of knowledge should fail on all the questions that need it, so its errors should cluster on related questions. We call the tendency of questions to be answered right or wrong together *cross-question structure*. Models' errors are related: different models often choose the same wrong answer (Kim et al., 2025), and a few latent factors summarize their benchmark scores (Burnell et al., 2023; Ruan et al., 2024; Maimon et al., 2026). We ask whether such structure comes from shared knowledge or from something else.

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Two other sources can produce it, and both are easy to miss. The first is the evaluation pipeline. Call the kth answers to all questions *pass* k; like one member of an ensemble, a pass answers every question once. Within one model, a knowledge gap makes a question hard in every pass, but it does not make two questions succeed or fail together from one pass to the next. Something shared by the questions of a pass can. If all questions of a pass share one random seed, they share the sampler's random numbers, so a lucky draw for the letter "B" in one question is, at the same generation step, the same lucky draw in all of them. Within one model, then, any structure comes from the pipeline, not from knowledge, which makes the within-model runs a control. The second source is a habit of the model. Llama-3.2-1B, for instance, picks "A" nine times in ten when it guesses (Fig. 1). It therefore looks as if it knows every question whose correct answer is A, and across an ensemble those questions go right or wrong together, which is what shared missing knowledge would look like. Such letter preferences are known for single models (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024; Ding, 2026). Missing knowledge should follow a question's content; a letter habit should follow the letter on which the answer is shown.

We test for both. Within one model, we compare runs that draw every answer independently with runs that share a seed or a few-shot prompt across questions (Sec. 3.1). Across 24 models, we move the answer options to new letters, which changes the correct letter but not the content (Sec. 3.2), and then count a question correct only if a model is right

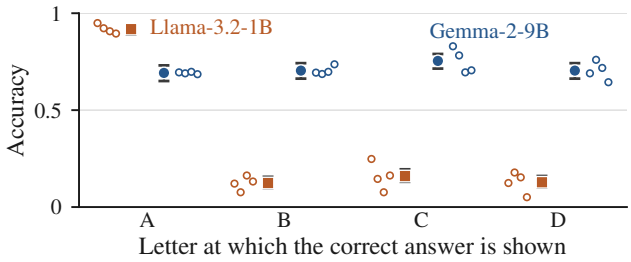


Figure 1. Llama-3.2-1B is accurate where the answer is shown at A, whatever the question; Gemma-2-9B is about equally accurate at every letter. Filled: all 500 questions, Wilson 95% CI. Open: each rotation alone (0–3, left to right), over the 118 to 129 questions whose answer it shows at that letter.

under every letter order (Sec. 3.3). Throughout, we measure structure as the excess of the eigenvalues of the question correlation matrix over what independent answers, or one skill per model, would produce. Independent sampling shows no structure that we can detect, and a shared seed or prompt creates it. Across models, the largest structure after skill is letter preference, mostly Llama-3.2-1B's; once letters are rotated, only a marginal excess remains, carried mostly by near-duplicate models. Spectral measures of benchmark dimensionality exist (Sha & Zhao, 2026), and the pull of few-shot exemplars is known (Zhao et al., 2021; Min et al., 2022). What we add is that a correctness matrix, which records no letters, still carries a letter habit, mostly one model's, as its second-largest direction (App. A).

2. A spectral test for cross-question structure

We write the answers as a correctness matrix $\mathbf{C}$: entry c_{ki} is 1 if row k answers question i correctly and 0 otherwise. Within one model a row is a pass; across models a row is a model. We keep only questions that vary: within a model, the N_b *borderline* questions, whose pass rate f_i (the share of passes that answer question i correctly) lies between $\varepsilon = 0.05$ and $1 - \varepsilon$; across models, the *split* questions, which some models answer correctly and others do not.

Let $\mathbf{R}$ be the correlation matrix of the kept questions. A group of questions that tend to be right or wrong together gives $\mathbf{R}$ a large eigenvalue. Its eigenvector, which we call a *direction*, says which questions belong to the group (each question's *loading*) and which rows follow it (each row's *score*). The question is how large is too large. If the questions were independent, the eigenvalues would fill the Marchenko–Pastur range, whose upper edge $\lambda_+ = (1 + \sqrt{\gamma})^2$ depends only on the ratio γ of questions to rows (Marčenko & Pastur, 1967).

Within one model, independence is the right reference: for a fixed model each question has its own chance of being answered correctly, and only the pipeline can tie questions together. We count the eigenvalues above λ_+ (the *MP count*) and compare the largest with the same eigenvalue in 1000 simulated matrices of independent answers at the observed pass rates; we call such a simulated reference a *null* (App. C, which also gives a Tracy–Widom test (Johnstone, 2001)).

Across models, independence is the wrong reference. A stronger model answers more questions correctly, so every question is tied to model skill. The reference is instead an item response model that gives each model m one skill θ_m and each question i a difficulty b_i and a slope a_i: the model answers correctly with probability $\sigma(a_i(\theta_m - b_i))$, with σ the logistic function (the two-parameter logistic, or 2PL, model (Lord & Novick, 1968)). One skill can already produce a second large eigenvalue with 0/1 answers, through a component that separates easy from hard questions, the *difficulty factor* of binary items (Ferguson, 1941; McDonald & Ahlawat, 1974); judged by λ_+, that would look like structure. So we fit the one-skill model, simulate datasets from it, and say that the jth largest eigenvalue λ_j (rank j) *exceeds* it if it lies above the 99th percentile of the simulated λ_j. We fit several variants of this null: without guessing (the 2PL, and Rasch, which gives every question the same slope (Rasch, 1960)) and with guessing (the three-parameter 3PL, with a guessing floor of 0.25, and know-or-guess variants; App. F), and keep a variant only if it is *valid*: if it reproduces the observed first eigenvalue, which measures how tightly skill ties the answers together. An excess can also spread over several eigenvalues, so a *family-wise test* asks whether the most extreme of the second to fourth eigenvalues is more extreme than in the simulations, with p-value p_M. We set the validity rule after our first analyses. We then wrote down the family-wise test and the *sibling groups* of near-duplicate models (versions or fine-tunes of one base model at one size) before running them, but after a post-hoc joint statistic over ranks 2 to 4, which rejected, the per-rank tests and the model-pair correlations had been computed on the same data, so these tests are not independent of those earlier analyses. Removals of single models or model pairs, and the size, power and difficulty analyses, are post hoc.

Rotations. Each model answers every question four times, with the options moved zero, one, two or three letters later, cyclically. The *gold letter*, the letter of the correct option, changes; the content does not. A pattern of content stays with the question, and a letter habit follows the letter. Counting a question correct only if it is right in all four rotations, which we call *all-4-correct* (the CircularEval rule of Liu et al. (2024)), leaves a model that guesses by letter at most a $1/256$ chance on any question, the same for every question.

Runs. Within one model (App. B), twelve *independent-draw runs* give every answer its own seed, as self-consistency does: five models on 500 MMLU ques-

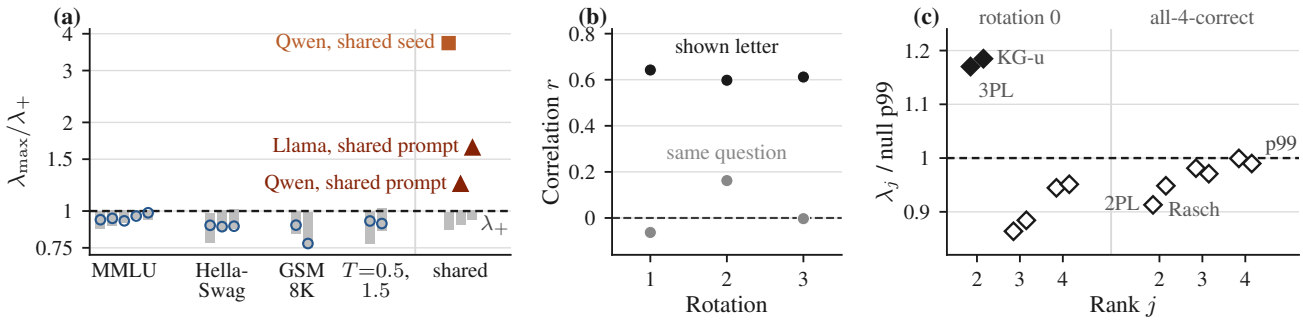


Figure 2. **(a)** Within one model: $\lambda_{\max}/\lambda_+$ (log axis) of each run; grey bar: 5–95% of its simulations; circles: independent draws; filled: an eigenvalue above λ_+. **(b)** Across models: correlation, over questions, of each question's eigenvector-2 loading in rotations 1–3 with the rotation-0 mean loading of the questions whose gold letter it now shows, and with its own rotation-0 loading. **(c)** λ_2–λ_4 of the 24-model matrix in rotation 0 and all-4-correct (Sec. 3.3), divided by the 99th percentile of two valid one-skill variants (App. F; rotation 0: 3PL left, KG-u, a know-or-guess model that guesses at random, right; all-4-correct: 2PL left, Rasch right); filled: above it.

tions (Hendrycks et al., 2021), three on HellaSwag (Zellers et al., 2019), Qwen2.5-7B on GSM8K (Cobbe et al., 2021) at two output lengths, and Qwen2.5-7B on MMLU at temperatures 0.5 and 1.5, each with $K = 30$ to 100 passes (Table A1). One *shared-seed run* gives all questions of a pass one seed, and two *shared-prompt runs* draw five MMLU development questions for each pass and place them, with their answers, before every question of that pass. Across models, 24 models from ten families answer the 500 MMLU questions in all four rotations by greedy decoding. Replies that cannot be read as one answer are scored wrong. They are at most 1.65% of replies within one model and 12.4% of one model's replies across models, but 217 of the 500 questions have at least one (App. B).

3. Results

3.1. Within one model: structure only from a shared component

In all 12 independent-draw runs, the MP count is zero: the largest eigenvalue stays below λ_+ and inside the range of its simulations (Fig. 2a; App. C). The test can see weak structure. A factor shared by the questions of a pass, strong enough to make their 0/1 answers correlate at about 0.025, is detected in every simulation at Llama-3-8B's 173 borderline questions, though only in 47% at Qwen2.5-7B's 43, and a cluster of nine questions can be missed (App. D).

Sharing a seed or a prompt changes this at once. In all three such runs the largest eigenvalue lies far above every simulation, and the MP count is two to four. The loadings show why. Our sampler (vLLM) divides each token's probability by a random number and emits the token with the largest ratio, so with a shared seed a letter that draws a small number at a given step is favored in every question of the pass that generates its letter at that step (App. E). In Qwen2.5-7B's shared-seed run, the gold letter accordingly explains a share $\eta^2 = 0.73$ of the variance of the top direction's loadings (permutation $p = 0.0002$). With a shared prompt, Llama-3-8B copies the answer letters of the exemplars, and its top direction also groups by gold letter. Qwen2.5-7B's does not, but two of its passes that happen to draw a common exemplar answer more alike than two that do not ($p = 0.0005$).

3.2. Across models: the second direction is letter preference

In rotation 0, the first direction of the 24-model matrix is skill: the models' scores on it correlate with their accuracy at $r = 0.94$. Here only one-skill variants that allow guessing are valid. The second eigenvalue, 49.9 (the edge λ_+ is 28.9 for the 460 split questions), exceeds the 99th percentile of each of the three valid variants that let a model guess at random (3PL and KG-u) or by its own letter preference (a *letter-aware* know-or-guess null), and it is the only eigenvalue that does; the family-wise test rejects all three ($p_M = 0.002$; Fig. 2c).

What is this second direction? Its loadings group by gold letter, and the models' scores on it follow their share of A answers ($r = 0.76$) and, with the opposite sign, of D answers ($r = -0.90$). Within one rotation, however, a question's gold letter is fixed, so this cannot tell a letter habit from content. Rotation can. In rotations 1 to 3 we correlate each question's loading with two quantities from rotation 0: its own loading, and the average loading of the questions that had its new gold letter. A content direction would follow the first, a letter habit the second. The direction follows the letter ($r = 0.60$ to 0.64) and hardly the question ($r = -0.06$ to 0.16; Fig. 2b), which leaves room for at most a small part that follows content. A fourth valid variant, the *soft-preference* null, adds to one skill a fitted letter preference that acts on every answer, not only on guesses. The family-wise test does not reject it ($p_M = 0.079$), but it makes the observed second eigenvalue unlikely ($p = 0.010$; App. F),

so even an explicit letter habit on top of one skill does not fully reproduce the direction.

Most of the direction is Llama-3.2-1B's habit. Removing it lowers the second eigenvalue from 49.9 to 35.0, below what one skill produces for the other 23 models, while removing any other single model leaves it between 47.1 and 52.4. A weaker excess then remains at the fourth eigenvalue, and the family-wise test still rejects the 3PL null ($p_M = 0.003$; App. F).

3.3. After rotation: a marginal excess, mostly from two model pairs

Counting a question correct only if it is right in all four rotations removes a letter habit that acts on guesses; whether a habit that also tilts answers a model partly knows survives, we cannot test (App. F). In the all-4-correct matrix, the second eigenvalue lies inside the range of the valid one-skill variants (Fig. 2c). That range is wide, because one skill alone already lifts the second eigenvalue above λ_+ through the difficulty factor. Could a real second skill hide in it? We planted one in simulation, on the third of the questions that come from STEM subjects and as strong as the first skill (the same spread across models and the same median slope). The second eigenvalue alone caught it in 69 to 78% of datasets, and the family-wise test in more than 90%. A second skill half as strong would usually be missed (App. F).

What remains is a marginal excess spread over the third and fourth eigenvalues. No single eigenvalue exceeds both valid variants (no all-4-correct marker in Fig. 2c is filled), but the family-wise test rejects them ($p_M = 0.026$ and 0.041; Table A2), though it is slightly liberal here (Sec. 4). Most of this excess comes from near-duplicate models. Keeping one model of each sibling group (Phi-3/3.5-mini, Llama-3/3.1-8B, Qwen2/2.5-7B, and Mistral-7B v0.2/v0.3 with Zephyr-7B) ends the rejection in all 24 ways of choosing, while removing as many other models at random never does. A follow-up, chosen after seeing this result, isolates two pairs: removing one model of Phi-3-mini/Phi-3.5-mini and one of Llama-3-8B/Llama-3.1-8B raises p_M above 0.2, more than any of 190 removals of two other models. Llama-3.2-1B carries part of the excess too: removing it alone also ends the rejection, though less clearly. Simple checks agree: letter shares flag Llama-3.2-1B among three letter-biased models, and residual correlations of the one-skill fit rank the two pairs first and second of 276 (App. F).

4. Discussion

In our data, the structure beyond skill that we can attribute comes from something other than knowledge: from a seed or prompt shared within one model, and from a letter habit across models, mostly Llama-3.2-1B's. After rotation, a marginal excess of untested origin remains. One habit of one model is enough to create a direction that an ensemble would read as missing knowledge, and scores that treat cross-model disagreement as epistemic (Hamidieh et al., 2026) should be checked for it. For self-consistency, a shared seed leaves the distribution of each question's answers unchanged but couples the questions of a pass, so it matters for any analysis that compares or pools answers across questions.

The remaining excess is marginal. Under one skill per model, the family-wise test rejects in up to 8.7% of simulated datasets at the 5% level, the calibrated p_M are 0.017 to 0.057, and the rejection ends when the 217 questions with an unparseable reply are dropped, so parse failures may carry part of the excess, though only 236 split questions then remain. The two pairs were chosen after seeing the data, and why they fail together, whether through knowledge gaps from shared training data, contamination or another cause, we did not test; architecture, family, scale and training data vary together in the ensemble. The test can miss a second skill half as strong as the first, structure carried by a few models, and anything outside correctness, such as log-probabilities. Our runs cover instruction-tuned models of 1.7B to 8B parameters within one model and 500 MMLU questions across models, and the thresholds are not corrected for the several variants and scores we tried. We did not test whether item-level factor and item response analyses (Kipnis et al., 2025; Sha & Zhao, 2026; Yao et al., 2026) or shared wrong answers (Kim et al., 2025) carry the same letter confound.

Before reading an ensemble's errors as missing knowledge, we recommend drawing every sample with its own seed, rotating the answer options and counting a question correct only if it is right in every rotation, testing what remains against a fitted one-skill model, and checking whether near-duplicate models carry it.

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A. Background

Which parts of the method are classical? The eigenvalue test and the one-skill models of educational testing are; what is new is the effect of a letter preference on a correctness matrix.

Eigenvalue tests. n questions with pairwise correlation κ have a population eigenvalue of $1 + (n-1)\kappa$, which grows with n, so an eigenvalue test can detect a correlation smaller than the noise of one pairwise correlation, about $1/\sqrt{K}$. The MP law holds for sample covariance matrices of independent entries with finite variance (Marčenko & Pastur, 1967; Bai & Silverstein, 2010) and for sample correlation matrices of i.i.d. entries (Jiang, 2004). For Gaussian data, the largest eigenvalue minus λ_+, divided by $\sigma_{\mathrm{TW}} = (1 + \sqrt{\gamma})(1 + 1/\sqrt{\gamma})^{1/3} K^{-2/3}$, converges to the Tracy–Widom law (Johnstone, 2001); this ratio is our z. For correlation matrices the limit is proved, as far as we know, only without centering or for Gaussian entries (Bao et al., 2012; Pillai & Yin, 2012); our columns are Bernoulli with unequal pass rates and centered at their sample means, so we calibrate z by simulation (App. C).

Option position. Ding (2026) shuffle the options and find that part of a model's self-agreement depends on option position. In the option-level response model of Fei et al. (2026), a per-model positional preference barely helps to predict held-out correctness (accuracy 0.840 with it, 0.838 without). Sec. 3.2 shows that such a preference can still create the second direction of a correctness matrix.

B. Data collection

Could the parser or the data explain the results? Within one model, no count changes when unparseable replies are dropped or imputed. Across models, dropping every question with an unparseable reply ends the marginal all-4-correct rejection.

Questions. MMLU: ten random questions from each of the first 50 subjects in alphabetical order. HellaSwag: 200 validation items, sampled within each activity label. GSM8K: the first 100 test problems. In the shared-prompt runs, each pass draws its own five exemplars from the 285 MMLU development questions (so two passes can share one) and shows them with every question; both runs use the same sets.

Prompts and decoding. Each model receives one user message in its own chat template (raw text for Phi-2, which has none), checked against an independent reference tokenization; shared-prompt messages place the five exemplars, with their answer letters, before the question. Generation uses vLLM 0.23.0 in bfloat16 with plain temperature sampling and at most 8 new tokens for MMLU and HellaSwag and 256 (or 1024) for GSM8K; the 24-model runs decode greedily with deterministic kernels. Every sample (k, i) has its own seed, except in the shared-seed run. vLLM gives each seeded request its own random-number generator, so its noise does not depend on its position in a batch.

Scoring. A multiple-choice reply is read as letter X when it begins with X (optionally after "Answer:", a parenthesis or bold markers) and does not negate or hedge it, or when its first line says "the answer is X" or similar. A reply that names several letters, negates its letter, or fits neither rule is unparseable and scored wrong. A GSM8K reply is correct when the number after its last "####", or else its last number, equals the gold answer.

Unparseable replies. In the within-model runs, at most 1.65% of replies are unparseable (Llama-3-8B on HellaSwag); in the ensemble, Mistral-7B v0.2 has 12.4% and every other model at most 9.6%. Dropping every question with an unparseable reply, or imputing each such reply by the question's mean, leaves every count of the independent-draw runs at zero and both counts of Llama-3-8B's shared-prompt run at two. On all-4-correct, the family-wise test still rejects after imputation and without the 95 questions whose score an unparseable reply decides, but not without all 217 questions that have one (Table A2). In rotation 0, dropping the 112 questions with an unparseable reply moves rank 2 to the 98.3rd percentile of the letter-aware null (App. F); it stays above the 3PL null, and the family-wise test rejects both ($p_M = 0.013$ and 0.004).

Truncation on GSM8K. At 256 new tokens, 55% of Qwen2.5-7B's replies are cut off, mostly before the final answer (13.6% of them correct, against 96.7% of complete ones; accuracy 51.0% at 256 tokens and 90.3% at 1024). Truncation changes which answers are correct but does not couple two questions of a pass, so the run still tests independence.

Table A1. The test on every within-model run ($\varepsilon = 0.05$). λ_+: MP edge; TW z: Tracy–Widom statistic. Counts: MP (above λ_+) / buffered (above $1.05\,\lambda_+$, a margin for the fluctuation of $\lambda_{\max}$). GSM8K runs use $T = 1$.

Run	N_b	λ_+	$\lambda_{\max}$	TW z	Counts
Independent, MMLU ($K = 100, T = 1$)					
Qwen2.5-7B	43	2.74	2.56	-1.72	0/0
Mistral-7B-v0.3	83	3.65	3.45	-1.79	0/0
Llama-3-8B	173	5.36	4.97	-3.05	0/0
SmolLM2-1.7B	406	9.09	8.74	-2.17	0/0
Phi-3-mini	271	7.00	6.91	-0.65	0/0
Independent, HellaSwag ($K = 50, T = 1$)					
Qwen2.5-7B	16	2.45	2.19	-1.60	0/0
Mistral-7B-v0.3	65	4.58	4.05	-2.71	0/0
Llama-3-8B	86	5.34	4.75	-2.90	0/0
Independent, Qwen2.5-7B ($K = 30$)					
GSM8K, 256 tok.	62	5.94	5.32	-2.06	0/0
GSM8K, 1024 tok.	19	3.22	2.50	-2.97	0/0
MMLU, $T = 0.5$	25	3.66	3.39	-1.08	0/0
MMLU, $T = 1.5$	85	7.20	6.53	-2.07	0/0
Shared component, MMLU ($K = 100, T = 1$)					
Qwen, seed	41	2.69	10.00	$+70.2$	4/4
Qwen, prompt	93	3.86	4.77	$+7.9$	2/2
Llama, prompt	264	6.89	11.31	$+30.9$	2/2

C. Calibration of the within-model test

Is the within-model test calibrated? No theorem we know of gives the limit of z for our matrix (App. A), so we judge z and the MP count against Monte Carlo nulls of independent answers. Against them, no independent-draw run shows structure at any borderline threshold, and every shared-component run does at all of them.

The null and z. The null treats each entry c_{ki}, given the model and the prompts, as a Bernoulli variable with $\Pr(c_{ki} = 1) = p_i$, independent across questions and passes; the borderline filter acts on each column alone, so the kept columns stay independent. At $K \leq 100$ the null distribution of z is shifted from the Tracy–Widom law: every z of the 12 independent-draw runs is negative, and their Monte Carlo nulls of z have means between -1.98 and -1.47. A z above every draw of its null, as in the three shared-component runs, rejects independence, but it does not say whether the dependence lies between questions or between passes.

Checks of independence. Three further checks agree with independence. The number of correct borderline answers per pass varies as independent draws predict (observed over predicted variance 0.61 to 1.09; no run overdispersed against a column-shuffle null, smallest $p = 0.27$; the two lowest ratios, $p = 0.039$ and 0.045 for underdispersion, are not significant after a Holm correction over 48 tests: 12 runs $\times$ over- and underdispersion and two trends over the pass index). This check cannot see a coupling that cancels in the number correct, such as the shared seed's (App. E). Borderline questions of one subject are no more correlated than other pairs (Qwen2.5-7B and Llama-3-8B, MMLU, $T = 1$: mean -0.018, $p = 0.80$, and 0.002, $p = 0.37$). No pair of borderline questions is coupled beyond its permutation null, in correctness for the five MMLU runs at $T = 1$ or in the answer letters of Qwen2.5-7B (largest Cramér's V 0.41, null 99th percentile 0.72).

Borderline threshold. At $\varepsilon = 0.01, 0.02, 0.05, 0.10, 0.15$ and 0.20, with the filter re-applied to the data and to each of the 1000 null matrices, no independent-draw run has an eigenvalue above its edge or a z above the 95th percentile of its null, and every shared-component run keeps at least two eigenvalues above its edge. The six tests of a run are nested, and at $\varepsilon = 0.20$ the three smallest sets have only 6, 7 and 10 questions.

D. Power of the within-model test

Could the within-model test miss weak structure? A factor shared by all borderline questions of a pass, with latent correlation 0.05, is found in every simulation at 173 borderline questions but in only 47% at 43, and a cluster of nine questions is usually missed at every strength we simulated.

Simulation. The matrices have the pass rates of two independent-draw MMLU runs, Qwen2.5-7B ($N_b = 43$) and Llama-3-8B ($N_b = 173$), $K = 100$ passes and all 500 questions. Entry c_{ki} is 1 when a hidden standard normal score falls below $\Phi^{-1}(f_i)$, so each question keeps its pass rate. On loaded questions the score is $\sqrt{\rho}\,u_k + \sqrt{1-\rho}\,e_{ki}$, with u_k shared by the loaded questions of pass k and e_{ki} independent noise; elsewhere it is e_{ki}. Two loaded questions then correlate at ρ in their scores and, at these pass rates and averaged over pairs, at about $\rho/2$ in their 0/1 entries. The factor loads on all borderline questions (diffuse) or on 30 or 9 chosen at random. A count of one or more, with the filter re-applied, detects the factor. Each scenario has 500 matrices (2000 at $\rho = 0$).

Results. A diffuse factor is easier to detect at Llama-3-8B's 173 borderline questions, because it loads four times as many (Fig. A1). A cluster of fixed size is harder to detect there, because it is a smaller share of them: a 9-question cluster at $\rho = 0.2$ is detected in 23% of Qwen2.5-7B and 11% of Llama-3-8B simulations. The false-positive rates are 4.6% and 8.3% for the MP count and 0.65% and 0.35% for the buffered count, which has less power (22% and 99.8% for a diffuse factor at $\rho = 0.05$). We did not simulate other values of N_b.

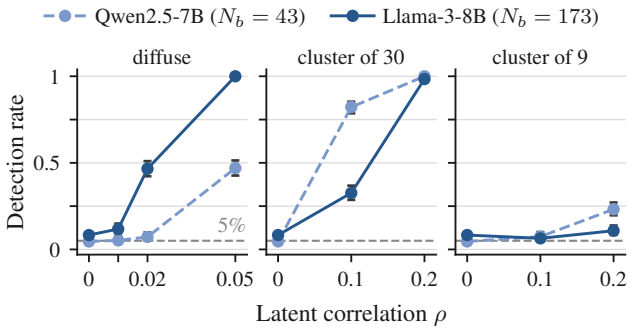


Figure A1. Detection rate of the MP count against the latent correlation ρ of a factor planted on all borderline questions or on 30 or 9 of them, at the pass rates of two MMLU runs. Whiskers: Wilson 95% intervals.

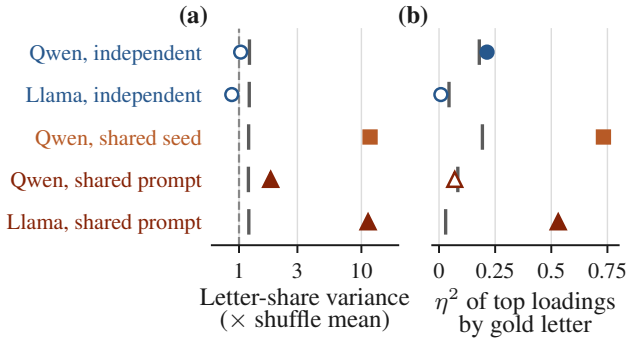


Figure A2. A shared seed or prompt makes a pass's letter shares vary beyond chance. Of the shared runs, only the seed run and Llama-3-8B's prompt run also group the top loadings by gold letter. **(a)** Across-pass variance of a pass's letter shares, summed over letters, divided by its mean when answers are shuffled across passes. **(b)** η^2 of the top eigenvector's loadings by gold letter. Grey tick: null 99th (a) or 95th (b) percentile; filled: beyond it.

E. What the shared component couples

Where does a shared seed or prompt couple the questions? A shared seed couples them through the sampler's noise for each letter; a shared prompt, through the exemplars' answer letters for Llama-3-8B and through the exemplar set itself for Qwen2.5-7B.

Shared noise. At each generation step, a seeded request draws one exponential random number, its noise, for every token and emits the token whose probability at temperature T, divided by its noise, is largest (Gumbel-max sampling), so each answer alone follows the model's distribution. Two requests with the same seed draw the same noise at every step, whatever their prompts, as coupled generation and counterfactual sampling use on purpose (Corvelo Benz et al., 2026; Ravfogel et al., 2025). If a shared-seed pass draws a small noise value for "B" at some step, "B" gains in every question whose letter is generated at that step.

The shared-seed run. In Qwen2.5-7B's shared-seed run, the across-pass variance of a pass's letter shares (its fractions of A, B, C and D answers), summed over letters, is 11.8 times its mean when each question's answers are shuffled across passes, against 1.03 in its independent-draw run (Fig. A2a). A pass whose noise favors one letter gains on questions with that gold letter and loses on the others, so the top loadings group by gold letter (Sec. 3.1), while the gains and losses cancel in the number correct per pass (observed over predicted variance 0.83).

The shared-prompt runs. Llama-3-8B copies the exemplars' letters, as few-shot prompts are known to make models do (Zhao et al., 2021): the number of a pass's exemplars whose answer is letter L correlates with its share of answers L at $r = 0.35$ to 0.47. Its letter shares vary 11.3 times their shuffle mean, and its number correct per pass is overdispersed (ratio 1.51, $p = 0.001$). Qwen2.5-7B's letter shares vary 1.8 times their shuffle mean but do not follow the exemplars' letters ($|r| \leq 0.09$; loadings by gold letter $\eta^2 = 0.07$). Its passes that share an exemplar still answer the borderline questions more alike than passes that share none, by 10.7 standard deviations of a null that permutes exemplar sets (5.9 for Llama-3-8B; $p = 0.0005$ for both). Links to the exemplars' content are weak (subject grouping $p = 0.042$, uncorrected; a same-subject exemplar adds 3.2 points of accuracy on borderline questions, $p = 0.033$). Neither Llama-3-8B's nor the shared-seed direction groups by subject ($p = 0.35$ and 0.23).

An unexplained value. Qwen2.5-7B's independent-draw run shows $\eta^2 = 0.21$ by gold letter ($p = 0.023$; Fig. A2b), although its $\lambda_{\max}$ lies below λ_+ and the answer-letter test of App. C finds no coupled pair. That test looks for single coupled pairs, not for a grouping of loadings. We do not interpret this value.

F. The 24-model ensemble

Do other nulls, scores or model removals change the across-model results? In rotation 0, every valid null whose letter preference, if any, acts only on guesses leaves the second eigenvalue in excess; the soft-preference null, whose preference acts on every answer, makes it unlikely but fits the data poorly. On all-4-correct, the marginal excess ends when one model of each of two pairs, Llama-3.2-1B (less clearly), or the 217 questions with an unparseable reply are removed.

F.1. Models, nulls and the family-wise test

Roster. Five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B and 7B; Qwen2-7B), five Mistral (Mistral-7B v0.2 and v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), four Llama (Llama-3.2 at 1B and 3B, Llama-3-8B, Llama-3.1-8B),

three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), two Gemma-2 (2B, 9B), and SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-LLM-7B and Granite-3.1-8B. Phi-2 is a base model; the other 23 are instruction-tuned. Rotation-0 accuracy runs from 35.0% (Llama-3.2-1B) to 72.8% (Gemma-2-9B).

One-skill nulls. In all of them skill is the only link between questions, and every draw is conditioned on each kept column staying split. The 2PL (Sec. 2) treats the skills as fixed and integrates the question parameters over a normal population (marginal maximum likelihood); each simulated column draws its (a_i, b_i) from its posterior, fitted on all 500 columns. Rasch is fitted by joint maximum likelihood. The 3PL is $0.25 + 0.75\,\sigma(a_i(\theta_m - b_i))$. In the uniform know-or-guess (KG-u) null, a model knows a question, with the 2PL probability, in all four rotations at once, or else guesses a letter uniformly in each rotation; it is fitted to all four rotations and scored like the data. The letter know-or-guess (KG-l) null guesses from the model's letter preference π_m instead; the letter-aware null is a know-or-guess model fitted to rotation 0 with π_m from all four rotations. The soft-preference (SP) null is in App. F.2. We also tried a Bock–Aitkin 2PL (skills integrated, slopes at most 4) and a fixed-margin null (uniform over 0/1 matrices with the observed margins).

Validity. A null counts only if the observed λ_1 lies between its 1st and 99th percentiles. In rotation 0 (1000 draws per null) it lies at the 76.4th percentile of the 3PL, 86.7th of the letter-aware, 98.7th of the KG-u and 85.7th of the SP null, which are valid, and at the 100th of the 2PL and Rasch. On all-4-correct (2000 draws) it lies at the 31.5th and 54.2nd percentiles of the 2PL and Rasch, which are valid, at the 100th of the 3PL and at the 0th of the KG-u and SP. KG-l, Bock–Aitkin and fixed-margin are invalid for both. A valid null's per-rank count is the number of ranks $2, 3, \ldots$ that each exceed its 99th percentile, counted until one does not; the rank-2 rule asks only about rank 2.

Family-wise test. The test uses 2000 new null draws, with seeds disjoint from all earlier runs. Its statistic M is the largest of the percentiles of λ_2, λ_3 and λ_4 among these draws, ties counted as half. Each draw's own M_d is computed against the other draws, and $p_M = (1 + \#\{d : M_d \geq M\})/2001$, at level 0.05. Three null draws reach $M = 100$, so p_M cannot fall below 0.002, the value in rotation 0.

Size. From each fitted one-skill model, 2PL and Rasch, we simulate 300 all-4-correct datasets, refit both nulls to each (500 draws), and apply the validity rule and the family-wise test (Table A2). The 2PL test rejects in 6.7% of datasets under 2PL truth and 8.7% under Rasch truth (95% intervals 4.1–10.1% and 5.7–12.4%), so it is not size-controlled by the bound of 7.5% we set before the run. The Rasch test rarely rejects under Rasch truth (0.7%), mainly because the refitted Rasch null is valid in only 23% of those datasets. The validity rule is thus not sufficient: it accepts the Rasch null on every 2PL dataset, despite wrong slopes, and rejects the correct Rasch model in most datasets drawn from it.

Other scores. A question can also count as correct in at least one, two or three rotations, or by the mean. On these scores some valid nulls are rejected (2PL on at least two, $p_M = 0.002$; Rasch, KG-l and 2PL on at least three, $p_M = 0.008$, 0.007, 0.009) and others are not (3PL and SP on at least two, $p_M = 0.86$ and 0.85; KG-u on at least three, 0.28), but data from the SP fit, with no second factor, make the refitted 2PL null reject in 98.5% and 58% of datasets, so we draw no conclusion. At least one of 4 gives $p_M = 0.996$; the mean has no valid null.

F.2. Rotation 0: the letter direction

Nulls with a letter preference. The letter-aware, 3PL and KG-u nulls let a letter preference act only on guesses; the SP null lets it also tilt answers a model partly knows. In 200 datasets simulated from the SP fit, which has no second factor, rank 2 exceeds the letter-aware null in 60% and the 3PL null in 99.5% (each null refitted, 200 draws), so under such a preference alone rank 2 often exceeds them.

The soft-preference null. Model m chooses option j of question i in a rotation with probability proportional to $\exp\big(b_{ij} + \mathbb{1}[j = g_i](a_i\theta_m + e_{mi})\big)\,\pi_m(L_j)$, where g_i is the gold option, b_{ij} a fitted intercept of option j with $b_{ig_i} = 0$, L_j the letter at which the rotation shows option j, and $e_{mi} \sim N(0, \tau^2)$ is shared by the four rotations of a cell and independent across models. Skill is the only link between questions, and π_m acts on every choice. The fit is penalized maximum likelihood over the parsed choices of all four rotations ($\tau = 1.17$). Simulations mask the cells unparseable in the data, and the rotation-0 result depends on this mask: the observed λ_2 lies at the 99.8th percentile of the null without it and at the 95.1st with it. Refitting inflates the slopes, so we calibrate: we simulate 200 datasets from the fit and refit each with its own null of 1000 draws; the calibrated p of rank j is the fraction of these datasets whose λ_j percentile is at least the observed one. In rotation 0 the calibrated p of λ_2 is 0.010 (0.015 by the convention $(1 + k)/(1 + n)$), and the family-wise test does not reject the SP null ($p_M = 0.079$). The SP null is invalid without Llama-3.2-1B and on all-4-correct, so we cannot test whether a preference that tilts partly known answers survives all-4-correct.

Predictive check. The SP model fails its predictive check, set before the run, against 500 datasets simulated from the fit. Per-model accuracy and letter shares fit, but the number of rotations in which a model answers a question correctly

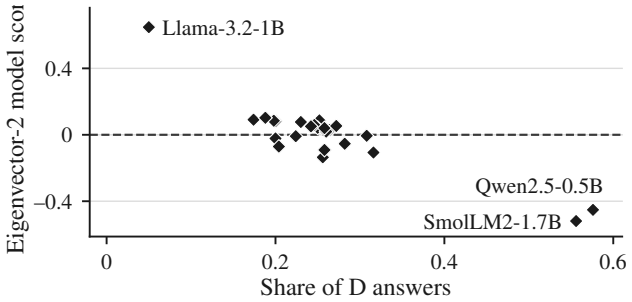


Figure A3. Each model's rotation-0 score on eigenvector 2 of the 24-model matrix against its share of D answers.

is covered in only 82.5% of model-by-count cells (90% required), and the observed number of split questions under "correct in at least two rotations" lies above all simulations. The weakest, letter-biased models have too many questions correct in exactly one rotation, and 21 of 24 models too many in all four (largest excess 0.06). The calibration is exact only if each refit inflates the slopes as much as the first, and it does not correct this misfit, so the SP verdicts are not fully validated.

Letter evidence. The models' scores on eigenvector 2 track their share of D answers ($r = -0.90$, permutation $p = 10^{-4}$; Fig. A3). Eigenvector 1 also carries a letter part ($\eta^2 = 0.22$ by gold letter), and eigenvectors 3 and 4 group by it more weakly but beyond chance ($\eta^2 = 0.087$ and 0.061, $p = 0.0002$ each). Because gold letters are nearly balanced (118 to 129 per letter), a letter preference leaves expected benchmark scores nearly unchanged.

Without Llama-3.2-1B. The 3PL and letter-aware nulls are both still rejected ($p_M = 0.003$ and 0.015), and the letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$). Ranks 3 and 4 lie at the 98.5th to 99.9th percentiles of these two nulls, and in their 1000 draws rank 4 exceeds the 99th percentiles by only 0.35 and 0.57.

F.3. All-4-correct: the marginal excess

Power. We simulate from the valid all-4-correct 2PL fit plus a second skill. As in the 2PL null, each of the 391 kept columns draws (a_i, b_i) from its posterior. On the 125 of them from the 18 STEM subjects, the logit gains $s\,\bar{a}\,\mathrm{sd}(\theta)\,z_m$, where $z_m \sim N(0,1)$ is independent of θ_m, $\bar{a} = 1.71$ is the median slope and $\mathrm{sd}(\theta) = 1$; at $s = 1$ the second skill matches the first in spread and median slope. We refit both nulls to each of 200 datasets per s (500 draws; rates in Table A2). At $s = 2$ the family-wise test and the rank-2 rule detect the second skill in at least 97% of datasets. The median λ_2 is 28.2 at $s = 0$, 28.3 at $s = 0.5$ and 33.8 at $s = 1$, against the observed 29.0.

Table A2. Family-wise test on all-4-correct (reject at $p_M < 0.05$). Subsets: number that reject, and the range or maximum of p_M. Pair removals: one Phi and one Llama model. Other two removed: the 190 removals of two other models (fraction that reject, maximum p_M). Unparseable: replies imputed, or questions dropped, either the 95 whose score they affect or all 217 that contain one (1000 draws). Size: rejection rate in 300 datasets simulated from each fitted one-skill model. Calibrated p_M: under 2PL and Rasch truth. Power: family-wise / rank-2 rule, 200 datasets per strength s (2PL fit).

	2PL	**Rasch**
Pct. of $\lambda_2, \lambda_3, \lambda_4$	71.8, 98.5, 99.1	94.6, 97.0, 98.6
p_M, 24 models	0.026	0.041
Sibling subsets	0/24 (0.09–0.65)	0/24 (0.12–0.69)
Random subsets	24/24 ($\leq$0.047)	24/24 ($\leq$0.046)
Pair removals	0/4 (0.22–0.69)	0/4 (0.32–0.38)
Other two removed	0.90 ($\leq$0.099)	0.74 ($\leq$0.133)
Unparseable: impute	0.019	0.042
drop 95	0.036	0.043
drop 217	0.23	0.12
Size, 2PL truth	0.067	0.067
Size, Rasch truth	0.087	0.007
Calibrated p_M	0.033, 0.047	0.057, 0.017
Power, $s = 0.5$	0.20 / 0.01	0.21 / 0.07
Power, $s = 1$	0.91 / 0.69	0.93 / 0.78

The difficulty factor. In the all-4-correct nulls, the eigenvector-2 loadings follow question difficulty: the observed $|r|$ between loading and difficulty, 0.57, lies just below the 2PL null's 5th percentile and inside the Rasch null's 5–95% range. In the rotation-0 3PL null, by contrast, the median $|r|$ is 0.09 (observed 0.04). The observed loadings group by subject ($\eta^2 = 0.21$, $p = 0.001$), but a cubic in difficulty explains 39% of their variance, and without that trend the grouping is at chance ($p = 0.24$).

Model families. On all-4-correct, the models' scores on eigenvector 4, below λ_+, group by family ($\eta^2 = 0.74$, $p = 0.006$); in rotation 0, same-family pairs correlate little more than other pairs (mean r 0.414 against 0.375, $p = 0.085$).

Sibling models. We chose the sibling groups (Sec. 2) from model metadata. Keeping one model per group gives $2 \cdot 2 \cdot 3 \cdot 2 = 24$ subsets of 19 models; none rejects (both nulls refitted, 1000 draws; both valid in every subset), while all 24 controls, which remove five of the 15 other models at random, reject (Table A2). The subsets overlap heavily, so this is one contrast, not 24 replicates, and the random removals are not matched in size to the group models and include models from the same developers (Phi-2, Qwen2.5 at 0.5B to 3B, Llama-3.2, Ministral-8B, Mistral-Small). In rotation 0 the letter direction does not depend on the siblings: the 3PL and letter-aware nulls are valid and reject in all 24 sibling subsets.

Table A3. Simple baselines. Top: rotation-0 letter shares of the three models whose largest share exceeds 0.5; no other model's largest share exceeds 0.35. Bottom: correlation of model pairs' residuals after the all-4-correct 2PL fit, with rank among 276 pairs (mean -0.04, standard deviation 0.08).

Model	A	B	C	D
Llama-3.2-1B	0.81	0.04	0.09	0.05
Qwen2.5-0.5B	0.23	0.06	0.12	0.58
SmolLM2-1.7B	0.07	0.20	0.17	0.56

Model pair	r	**Rank**
Phi-3-mini, Phi-3.5-mini	0.40	1
Llama-3-8B, Llama-3.1-8B	0.27	2
Mistral-7B v0.2, v0.3	0.20	3
Mistral-7B v0.3, Llama-3-8B	0.19	4
Mistral-7B v0.2, Zephyr-7B	0.17	5
Qwen2-7B, Qwen2.5-7B	0.03	37

The two pairs. Removing one model of each of the two pairs of Sec. 3.3, in all four ways, gives a larger p_M under both nulls than every one of the 190 removals of two other models, of which 90% (2PL) and 74% (Rasch) still reject (1000 draws; every null valid; Table A2). The Qwen pair is not part of the excess (removals that include Qwen2-7B or Qwen2.5-7B reject more strongly, mean p_M 0.010 and 0.006 under the 2PL), nor is the Mistral group (removing v0.2 and v0.3, the residual rank-3 pair of Table A3, keeps $p_M = 0.019$; the other two removals in the group give 0.004 to 0.023), nor accuracy (the 15 removals of two models with rotation-0 accuracy 0.62 to 0.71, around the pair models' 0.644 to 0.684, give p_M at most 0.052 (2PL) and 0.113 (Rasch), below all pair removals).

Llama-3.2-1B. Removing Llama-3.2-1B alone gives $p_M = 0.059$ (2PL) and 0.089 (Rasch) and makes the KG-u null valid ($p_M = 0.15$); the sum $\lambda_2 + \lambda_3 + \lambda_4$, a secondary statistic fixed with the test but not used for decisions, still gives $p = 0.023$ and 0.016. Of the 19 of those 190 removals that include it, 12 (2PL) and 6 (Rasch) still reject. Removing it together with one model of each pair gives p_M 0.30 to 0.59 and 0.38 to 0.60, close to the pair removals.

Simple baselines. Rotation-0 letter shares flag three models whose most frequent letter exceeds half of their replies (Table A3); these shares count unparseable replies, whereas the preference of 0.90 for A in Sec. 1 is estimated over guesses. Correlations of model pairs' residuals (observed minus 2PL-expected correctness over the 391 all-4-correct columns) rank the Phi and Llama pairs first and second of 276.

F.4. Replicates, near ties and option references

Replicates. We collected the ensemble twice to test GPU nondeterminism and report replicate A. Replicate B differs in 6 of 48,000 chosen options and in eigenvalues by at most 0.081 over all matrices; in B alone, the all-4-correct λ_4 lies just above the 2PL rank-4 99th percentile of 1000 draws (20.82 against 20.78). A near tie is a cell whose two most probable first-token letters differ by under 0.13 nats in either replicate (5.0% of cells). Replacing near ties by the majority answer over rotations changes no verdict; without the 297 questions with a rotation-0 near tie (203 remain), rank 2 exceeds the 3PL but not the letter-aware null (both valid). Removing the 23 questions whose options refer to other options changes no per-rank count of rotation 0 or all-4-correct.